

Ziad Ashraf

Cairo, Egypt | 01069871230 | Ziadashraf98765@gmail.com | [GitHub](#) | [LinkedIn](#) | [Kaggle](#) | [BI-Portfolio](#)

AI Engineer with 2+ years of experience applying **Data Science**, **AI** and **ML** to solve real-world business problems. Hands-on experience in predictive modelling, NLP, Speech AI, deep learning, and end-to-end AI system development. Skilled in taking AI solutions from research and prototyping to production using modern **software engineering** principles, **MLOps**, and cloud-based deployment workflows.

Experience

Elshayeb

AI Engineer

Feb 2026 – Present

Alexandria | Onsite

- Architected end-to-end **AI solutions** for **media operations** and **CRM** workflows, automating manual processes and significantly reducing task completion time.
- Deployed production AI services using (**FastAPI**, **Docker**, **Nginx**, **AWS**) with **async REST APIs**, **caching**, and **rate limiting** (**Redis**), backed by **CI/CD pipelines** (**GitHub Actions**, **PyTest**, **Locust**) for reliable, fault-tolerant releases.
- Built **MLOps/LLMOps** infrastructure for **experiment tracking**, **prompt/model/ data versioning**, **cost optimization**, **monitoring** (**Prometheus**, **Grafana**, **Loki**), **observability** (**MLflow**, **W&B**, **Langfuse**), improving deployment confidence across AI services.
- Built an **Intelligent lead qualification** service using **Gemini** to generate tenant-specific onboarding questionnaires, incorporating token management, prompt versioning using **Langfuse**, and context-aware question generation for improved lead profiling.
- Designed a **two-stage multi-tenant SaaS lead retrieval system** leveraging **hybrid search** (**BM25 + dense embeddings**), **reranking**, and a custom relevance scoring algorithm to match lead profile answers with tenant-specific questionnaires, improving retrieval accuracy and search quality.
- Developed an **Egyptian ASR** plugin for podcast and reel transcription, combining **VAD**, **CTC models**, **multimodal processing**, and **LLM-based transcription** to achieve less than **7% WER** and **97% alignment accuracy**, reducing manual editing effort through Adobe Premiere Pro integration.
- Implemented a **Gemini-powered sales analytics** and **TTS** reporting service that analyzes sales representative performance, identifies short-term trends, and generates automated voice reports, reducing sales managers' time spent reviewing large teams.

ISFP

BI / ETL Developer

Jul 2026 – Present

Alexandria | Remote

- Delivering BI solutions that enable operational visibility and performance monitoring for government entities in the ports and logistics sector.**
- Developing** ETL workflows and data integration pipelines using **Alteryx**.
- Analyzing** structured and unstructured data using **Tableau**, including time series forecasting, to support operational reporting and business decision-making.
- Building and deploying** interactive dashboards on **Tableau Server** for reporting and operational monitoring.

SimpliLearn Education Consulting Private Limited

Data Scientist

Dec 2022 – Jun 2024

Alexandria | Remote

- Delivered **15+** data science projects, processing an average of **50K** records per project, attaining a **95%** client acceptance rate.
- Optimized **5+** ML models using **Python** and **R**, achieving a **5–30%** reduction in error, through hyperparameter tuning and feature engineering.
- Built **10+** interactive dashboards using **Power BI**, **Tableau**, and **Excel**, significantly increasing student engagement through dynamic data exploration.
- Conducted statistical analysis on **8+** datasets using **Python**, **SQL**, **SAS**, and **R**, ensuring **100%** timely delivery while increasing learner retention by **25%**.

Freelancer.com

Data Scientist

Jun 2022 – Apr 2022

Alexandria | Remote

- Deployed ML system on **AWS EC2** with automated retraining pipeline using **Python (scikit-learn)** and **Linux** (Cron jobs), achieving **100%** error-free deployment and reducing manual review time by **85%**.
- Cut annotation time by **90%** using point-supervised semantic segmentation on aerial images with **PyTorch (UNet)** and a partial focal loss function.
- Segmented **100M+** customers with **Spark** K-Means, revealing **97%** checkout abandonment and reducing training from **40** to **15** minutes with PCA.
- Applied **LightGBM** for **time series** demand forecasting across 75+ zones (4.5M+ monthly records, 20+ features), achieving a **60%** MAE reduction.
- Implemented 2 real-time IoT **ETL** pipelines and dashboards in **Zoho Analytics** to monitor the daily efficiency of **50+** devices, which reduced data errors by **90%** and cut reporting time by **70%** compared to manual CRM exports.
- Validated 3 research studies in **SPSS**, achieving over **90%** reliability, which supported evidence-based policy decisions.
- Received 5-star ratings from 8 clients for delivering high-quality, on-time results with comprehensive documentation and responsive support.

Projects

ContentLens AI: AI-Powered Marketing Document Platform

Dec 2025 – Feb 2026

Tech Stack: **Python**, **FastAPI**, **LangChain**, **LangGraph**, **Langfuse**, **Ollama**, **Cohere**, **AWS EC2**, **React**, **TypeScript**, **GitHub Actions**, **Tesseract**

- Architected **multi-agent system** with 9 **LangGraph**-orchestrated agents executing in parallel (6 concurrent), reducing **P90 latency** by **~70%** to **sub-3minute** response times under load.
- Engineered intelligent agents with automatic **Ollama-to-Cohere** fallback logic, achieving **82.5%** extraction quality and **99%** workflow reliability.
- Built **FastAPI** backend with **OCR** support (**Tesseract**), multilingual translation, and multi-format handling (PDF, DOCX, images, TXT) with async request processing, rate limiting, and retry mechanisms.
- Implemented **Langfuse** observability for **real-time** performance monitoring, **CI/CD** with **GitHub Actions** and **PyTest** coverage, and deployed cost optimized AWS infrastructure with **zero** LLM API costs.

Real Estate Intelligence System

Oct 2025 – Dec 2025

Tech Stack: **Python**, **Flask**, **Docker**, **AWS EC2**, **GitHub Actions**, **Hugging Face**, **Milvus**, **Prometheus**, **Grafana**, **Dagster**, **BigQuery**, **Power BI**

- Built a **Dockerized** end-to-end web scraping and ETL pipeline on **AWS** using **Dagster**, ingesting and cleaning **9K+ property records** from multiple Egyptian real estate websites into **BigQuery**.
- Preprocessed Arabic property ads and evaluated **Hugging Face** embedding models using **NDCG/MAP**, indexing embeddings in **Milvus** with **Pydantic validation** for scalable semantic search, achieving **92% data quality**.

- Deployed a **Flask-based RAG** system on **AWS** using **multilingual e5** embeddings model and vector search, delivering personalized AI broker recommendations with **<1s** search latency and **2–4s** end-to-end response time.
- Designed a six-page **Power BI** dashboard highlighting key investment metrics (**2.7%** monthly rent-to-price ratio for 3-year payback in Alexandria).

AI-Powered Mental Health Analysis System

Oct 2022 – Nov 2025

Tech Stack: Python, FastAPI, Docker, AWS, GitHub Actions, PostgreSQL, Airflow, MLflow, Hugging Face, PyTorch, TensorFlow, scikit-learn

- Built **FastAPI** microservices using **Docker** on **AWS (ECS, EC2, RDS)** with **GitHub Actions CI/CD** pipelines, reducing API response time from **0.8s to 0.5s**.
- Orchestrated 3 ML training pipelines with **Airflow**, implementing data versioning and model registration using **S3**, **MLflow**, and **WandB**.
- Developed sentiment classification models using **scikit-learn** and improved depression detection performance (**TPR**) by **31.9%** through **BERT-based LoRA fine-tuning** via **Hugging Face** and **PyTorch** for early detection of cases.
- Trained and fine-tuned 2 **CNN models** for facial recognition on 20K+ augmented images using **TensorFlow**, achieving real-time inference in **<2s**.

Skills

- **Programming & Tools:** Python, SQL, R, Git/GitHub, Linux
- **ML/DL Frameworks:** PyTorch, TensorFlow, scikit-learn, XGBoost, LightGBM, OpenCV
- **Machine Learning:** Regression, Classification, Time Series, Clustering, Ensemble Methods, A/B Testing
- **Deep Learning:** CNNs, Transfer Learning, Fine-tuning, YOLO, UNet, OCR
- **NLP & GenAI:** LLMs, RAG, AI Agents, Transformers, BERT, LoRA/PEFT, Hugging Face, LangChain, Embeddings, CrewAI, LangGraph
- **MLOps & Cloud:** Docker, Kubernetes, FastAPI, Flask, MLflow, CI/CD, AWS, GCP, GitHub Actions, Prometheus, Grafana, Langfuse
- **Databases:** PostgreSQL, MySQL, MongoDB, Qdrant, Milvus, BigQuery
- **Data Engineering:** Airflow, Dagster, Spark, Kafka, DVC, dbt, ETL/ELT, Web Scraping

Education

Alexandria University, Egypt

Sep 2019 – Jun 2023

Bachelor of Data Science and Computing